

Week 5 Assignment

Apache Spark DataFrames - Data Cleaning, Transformation and Aggregation

Name: Tanmay Mani Tripathi

Technology: PySpark

Platform: Google Colab

```
!apt-get install openjdk-11-jdk-headless -qq > /dev/null

!wget -q https://archive.apache.org/dist/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz

!tar -xzf spark-3.5.1-bin-hadoop3.tgz

!pip install -q pyspark

import os

os.environ["JAVA_HOME"]="/usr/lib/jvm/java-11-openjdk-amd64"
os.environ["SPARK_HOME"]="/content/spark-3.5.1-bin-hadoop3"

from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("Week5 Assignment") \
    .getOrCreate()

from google.colab import files

uploaded = files.upload()

<IPython.core.display.HTML object>

Saving Sample - Superstore.csv to Sample - Superstore (3).csv

import pandas as pd

pdf = pd.read_csv(
    "Sample - Superstore.csv",
    encoding="latin1"
)

pdf.head()

{"type": "dataframe", "variable_name": "pdf"}
```

```
df = spark.createDataFrame(pdf)
```

```
df.printSchema()
```

```
root
```

```
|-- Row ID: long (nullable = true)
|-- Order ID: string (nullable = true)
|-- Order Date: string (nullable = true)
|-- Ship Date: string (nullable = true)
|-- Ship Mode: string (nullable = true)
|-- Customer ID: string (nullable = true)
|-- Customer Name: string (nullable = true)
|-- Segment: string (nullable = true)
|-- Country: string (nullable = true)
|-- City: string (nullable = true)
|-- State: string (nullable = true)
|-- Postal Code: long (nullable = true)
|-- Region: string (nullable = true)
|-- Product ID: string (nullable = true)
|-- Category: string (nullable = true)
|-- Sub-Category: string (nullable = true)
|-- Product Name: string (nullable = true)
|-- Sales: double (nullable = true)
|-- Quantity: long (nullable = true)
|-- Discount: double (nullable = true)
|-- Profit: double (nullable = true)
```

```
df.show(5)
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+
|Row ID|      Order ID|Order Date| Ship Date|      Ship Mode|Customer
ID|  Customer Name|  Segment|      Country|      City|      State|
Postal Code|Region|      Product ID|      Category|Sub-Category|
Product Name|  Sales|Quantity|Discount|  Profit|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+
|      1|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|  CG-
12520|  Claire Gute| Consumer|United States|      Henderson|
Kentucky|      42420| South|FUR-B0-10001798|      Furniture|
Bookcases|Bush Somerset Col...| 261.96|      2|      0.0| 41.9136|
|      2|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|  CG-
12520|  Claire Gute| Consumer|United States|      Henderson|
```

```

Kentucky|      42420| South|FUR-CH-10000454|      Furniture|
Chairs|Hon Deluxe Fabric...|  731.94|      3|      0.0| 219.582|
|      3|CA-2016-138688| 6/12/2016| 6/16/2016|  Second Class|  DV-
13045|Darrin Van Huff|Corporate|United States|      Los Angeles|
California|      90036|  West|OFF-LA-10000240|Office Supplies|
Labels|Self-Adhesive Add...|   14.62|      2|      0.0|  6.8714|
|      4|US-2015-108966|10/11/2015|10/18/2015|Standard Class|  SO-
20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|
Florida|      33311| South|FUR-TA-10000577|      Furniture|
Tables|Bretford CR4500 S...|957.5775|      5|      0.45|-383.031|
|      5|US-2015-108966|10/11/2015|10/18/2015|Standard Class|  SO-
20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|
Florida|      33311| South|OFF-ST-10000760|Office Supplies|
Storage|Eldon Fold 'N Rol...|   22.368|      2|      0.2|  2.5164|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+
only showing top 5 rows

```

Q1. Why is Spark faster than MapReduce?

Spark performs in-memory processing whereas MapReduce writes intermediate results to disk after every stage. Spark is therefore much faster and suitable for iterative processing, machine learning, and real-time analytics.

Q2. What is Spark DataFrame?

A Spark DataFrame is a distributed table having rows and columns. DataFrames are immutable, meaning every transformation creates a new DataFrame.

Q3 Remove Duplicates

```

df_no_duplicates = df.dropDuplicates()

print("Rows Before:", df.count())
print("Rows After:", df_no_duplicates.count())

Rows Before: 9994
Rows After: 9994

```

Q4 Handle Null Values

```
df.na.drop().show(5)
```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+
|Row ID|      Order ID|Order Date| Ship Date|      Ship Mode|Customer
ID|  Customer Name|  Segment|      Country|      City|      State|
Postal Code|Region|      Product ID|      Category|Sub-Category|
Product Name|  Sales|Quantity|Discount|  Profit|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+
|      1|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|  CG-
12520|  Claire Gute| Consumer|United States|  Henderson|
Kentucky|      42420| South|FUR-BO-10001798|  Furniture|
Bookcases|Bush Somerset Col...| 261.96|      2|      0.0| 41.9136|
|      2|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|  CG-
12520|  Claire Gute| Consumer|United States|  Henderson|
Kentucky|      42420| South|FUR-CH-10000454|  Furniture|
Chairs|Hon Deluxe Fabric...| 731.94|      3|      0.0| 219.582|
|      3|CA-2016-138688| 6/12/2016| 6/16/2016|  Second Class|  DV-
13045|Darrin Van Huff|Corporate|United States|  Los Angeles|
California|      90036| West|OFF-LA-10000240|Office Supplies|
Labels|Self-Adhesive Add...| 14.62|      2|      0.0| 6.8714|
|      4|US-2015-108966|10/11/2015|10/18/2015|Standard Class|  S0-
20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|
Florida|      33311| South|FUR-TA-10000577|  Furniture|
Tables|Bretford CR4500 S...|957.5775|      5|      0.45|-383.031|
|      5|US-2015-108966|10/11/2015|10/18/2015|Standard Class|  S0-
20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|
Florida|      33311| South|OFF-ST-10000760|Office Supplies|
Storage|Eldon Fold 'N Rol...| 22.368|      2|      0.2| 2.5164|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+

```

only showing top 5 rows

```

from pyspark.sql.functions import *

df.select([
    count(when(col(c).isNull(), c)).alias(c)
    for c in df.columns
]).show()

```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+

```

```

+-----+-----+-----+-----+-----+-----+-----+
+-----+
|Row ID|Order ID|Order Date|Ship Date|Ship Mode|Customer ID|Customer
Name|Segment|Country|City|State|Postal Code|Region|Product ID|
Category|Sub-Category|Product Name|Sales|Quantity|Discount|Profit|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|      0|      0|      0|      0|      0|      0|      0|
0|      0|      0|      0|      0|      0|      0|      0|
0|      0|      0|      0|      0|      0|      0|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+

```

Q5 Filter Region

```
df.filter(col("Region")=="West").show()
```

```

+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+
|Row ID|      Order ID|Order Date| Ship Date|      Ship Mode|Customer
ID|      Customer Name| Segment|      Country|      City|
State|Postal Code|Region|      Product ID|      Category|Sub-Category|
Product Name| Sales|Quantity|Discount| Profit|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+
|      3|CA-2016-138688| 6/12/2016| 6/16/2016| Second Class| DV-
13045| Darrin Van Huff|Corporate|United States| Los Angeles|
California|      90036| West|OFF-LA-10000240|Office Supplies|
Labels|Self-Adhesive Add...| 14.62|      2|      0.0| 6.8714|
|      6|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-
11710| Brosina Hoffman| Consumer|United States| Los Angeles|
California|      90032| West|FUR-FU-10001487| Furniture|
Furnishings|Eldon Expressions...| 48.86|      7|      0.0| 14.1694|
|      7|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-
11710| Brosina Hoffman| Consumer|United States| Los Angeles|
California|      90032| West|OFF-AR-10002833|Office Supplies|
Art| Newell 322|      7.28|      4|      0.0| 1.9656|
|      8|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-

```

11710| Brosina Hoffman| Consumer|United States| Los Angeles|
 California| 90032| West|TEC-PH-10002275| Technology|
 Phones|Mitel 5320 IP Pho...| 907.152| 6| 0.2| 90.7152|
 | 9|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-
 11710| Brosina Hoffman| Consumer|United States| Los Angeles|
 California| 90032| West|OFF-BI-10003910|Office Supplies|
 Binders|DXL Angle-View Bi...| 18.504| 3| 0.2| 5.7825|
 | 10|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-
 11710| Brosina Hoffman| Consumer|United States| Los Angeles|
 California| 90032| West|OFF-AP-10002892|Office Supplies|
 Appliances|Belkin F5C206VTEL...| 114.9| 5| 0.0| 34.47|
 | 11|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-
 11710| Brosina Hoffman| Consumer|United States| Los Angeles|
 California| 90032| West|FUR-TA-10001539| Furniture|
 Tables|Chromcraft Rectan...|1706.184| 9| 0.2| 85.3092|
 | 12|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-
 11710| Brosina Hoffman| Consumer|United States| Los Angeles|
 California| 90032| West|TEC-PH-10002033| Technology|
 Phones|Konftel 250 Confe...| 911.424| 4| 0.2| 68.3568|
 | 14|CA-2016-161389| 12/5/2016|12/10/2016|Standard Class| IM-
 15070| Irene Maddox| Consumer|United States| Seattle|
 Washington| 98103| West|OFF-BI-10003656|Office Supplies|
 Binders|Fellowes PB200 Pl...| 407.976| 3| 0.2|132.5922|
 | 18|CA-2014-167164| 5/13/2014| 5/15/2014| Second Class| AG-
 10270| Alejandro Grove| Consumer|United States| West Jordan|
 Utah| 84084| West|OFF-ST-10000107|Office Supplies| Storage|
 Fellowes Super St...| 55.5| 2| 0.0| 9.99|
 | 19|CA-2014-143336| 8/27/2014| 9/1/2014| Second Class| ZD-
 21925|Zuschuss Donatelli| Consumer|United States|San Francisco|
 California| 94109| West|OFF-AR-10003056|Office Supplies|
 Art| Newell 341| 8.56| 2| 0.0| 2.4824|
 | 20|CA-2014-143336| 8/27/2014| 9/1/2014| Second Class| ZD-
 21925|Zuschuss Donatelli| Consumer|United States|San Francisco|
 California| 94109| West|TEC-PH-10001949| Technology|
 Phones|Cisco SPA 501G IP...| 213.48| 3| 0.2| 16.011|
 | 21|CA-2014-143336| 8/27/2014| 9/1/2014| Second Class| ZD-
 21925|Zuschuss Donatelli| Consumer|United States|San Francisco|
 California| 94109| West|OFF-BI-10002215|Office Supplies|
 Binders|Wilson Jones Hang...| 22.72| 4| 0.2| 7.384|
 | 25|CA-2015-106320| 9/25/2015| 9/30/2015|Standard Class| EB-
 13870| Emily Burns| Consumer|United States| Orem|
 Utah| 84057| West|FUR-TA-10000577| Furniture| Tables|
 Bretford CR4500 S...| 1044.63| 3| 0.0|240.2649|
 | 26|CA-2016-121755| 1/16/2016| 1/20/2016| Second Class| EH-
 13945| Eric Hoffmann| Consumer|United States| Los Angeles|
 California| 90049| West|OFF-BI-10001634|Office Supplies|
 Binders|Wilson Jones Acti...| 11.648| 2| 0.2| 4.2224|
 | 27|CA-2016-121755| 1/16/2016| 1/20/2016| Second Class| EH-
 13945| Eric Hoffmann| Consumer|United States| Los Angeles|

```

California|      90049|  West|TEC-AC-10003027|      Technology|
Accessories|Imation 8GB Mini ...|    90.57|      3|    0.0| 11.7741|
|      43|CA-2016-101343| 7/17/2016| 7/22/2016|Standard Class|    RA-
19885|      Ruben Ausman|Corporate|United States|    Los Angeles|
California|      90049|  West|OFF-ST-10003479|Office Supplies|
Storage|Eldon Base for st...|    77.88|      2|    0.0|    3.894|
|      63|CA-2015-135545|11/24/2015|11/30/2015|Standard Class|    KM-
16720|      Kunst Miller| Consumer|United States|    Los Angeles|
California|      90004|  West|TEC-AC-10004633|      Technology|
Accessories|Verbatim 25 GB 6x...|    13.98|      2|    0.0|    6.1512|
|      64|CA-2015-135545|11/24/2015|11/30/2015|Standard Class|    KM-
16720|      Kunst Miller| Consumer|United States|    Los Angeles|
California|      90004|  West|OFF-BI-10001078|Office Supplies|
Binders|Acco PRESSTEX Dat...|    25.824|      6|    0.2|    9.3612|
|      65|CA-2015-135545|11/24/2015|11/30/2015|Standard Class|    KM-
16720|      Kunst Miller| Consumer|United States|    Los Angeles|
California|      90004|  West|OFF-PA-10003892|Office Supplies|
Paper|      Xerox 1943|    146.73|      3|    0.0|   68.9631|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+
only showing top 20 rows

```

Q7 Total Sales

```

from pyspark.sql.functions import sum

df.select(
    sum("Sales").alias("Total Sales")
).show()

+-----+
|      Total Sales|
+-----+
|2297200.8602999896|
+-----+

```

Q8 Average Sales

```

from pyspark.sql.functions import avg

df.select(
    avg("Sales").alias("Average Sales")
).show()

```

```
+-----+
|      Average Sales|
+-----+
|229.85800083049725|
+-----+
```

Q9 Minimum & Maximum Sales

```
from pyspark.sql.functions import min,max

df.select(
    min("Sales").alias("Minimum Sales"),
    max("Sales").alias("Maximum Sales")
).show()
```

```
+-----+-----+
|Minimum Sales|Maximum Sales|
+-----+-----+
|          0.444|        22638.48|
+-----+-----+
```

Q10 Total Sales by Category

```
df.groupBy("Category") \
    .agg(
        sum("Sales").alias("Total Sales")
    ).show()
```

```
+-----+-----+
|      Category|      Total Sales|
+-----+-----+
|Office Supplies|719047.0320000007|
|      Furniture| 741999.7953000001|
|      Technology| 836154.0329999999|
+-----+-----+
```

Q11 Number of Orders by Region

```
from pyspark.sql.functions import count

df.groupBy("Region") \
    .agg(
        count("*").alias("Orders")
    ).show()
```

```

+-----+-----+
| Region|Orders|
+-----+-----+
|   South|   1620|
|Central|   2323|
|   East|   2848|
|   West|   3203|
+-----+-----+

```

Q12 Average Sales by Category in West Region

```

from pyspark.sql.functions import avg

df.filter(col("Region")== "West") \
.groupBy("Category") \
.agg(
avg("Sales").alias("Average Sales")
).show()

```

```

+-----+-----+
|      Category|      Average Sales|
+-----+-----+
|Office Supplies| 116.422376910912|
|      Furniture| 357.3023246110322|
|      Technology|420.68753255425725|
+-----+-----+

```

Q13 Rename Column

```

df2=df.withColumnRenamed(
"Customer Name",
"Customer_Name"
)

df2.printSchema()

root
|-- Row ID: long (nullable = true)
|-- Order ID: string (nullable = true)
|-- Order Date: string (nullable = true)
|-- Ship Date: string (nullable = true)
|-- Ship Mode: string (nullable = true)
|-- Customer ID: string (nullable = true)
|-- Customer_Name: string (nullable = true)
|-- Segment: string (nullable = true)
|-- Country: string (nullable = true)
|-- City: string (nullable = true)

```

```
| -- State: string (nullable = true)
| -- Postal Code: long (nullable = true)
| -- Region: string (nullable = true)
| -- Product ID: string (nullable = true)
| -- Category: string (nullable = true)
| -- Sub-Category: string (nullable = true)
| -- Product Name: string (nullable = true)
| -- Sales: double (nullable = true)
| -- Quantity: long (nullable = true)
| -- Discount: double (nullable = true)
| -- Profit: double (nullable = true)
```

Q14 Wide Transformation

Wide Transformation

Wide transformations require data to move between partitions. Operations like `groupBy()`, `join()`, and `distinct()` cause shuffle, which increases execution time because data is exchanged across executors.

Add blockquote

Q15 Complete Pipeline

```
pipeline = (
df.dropDuplicates()
.na.drop()
.groupBy("Category")
.agg(
sum("Sales").alias("Total Sales")
)
)

pipeline.show()

+-----+-----+
|      Category|      Total Sales|
+-----+-----+
|Office Supplies|719047.0320000031|
|      Furniture| 741999.795300001|
|      Technology|836154.0329999982|
+-----+-----+
```

Final Conclusion

Conclusion

In this assignment, Apache Spark DataFrames were used to perform data cleaning, filtering, aggregation, grouping, and schema modification. Spark's in-memory processing makes it significantly faster than Hadoop MapReduce and suitable for large-scale analytics.